

BCAC601, Unix and Shell programming

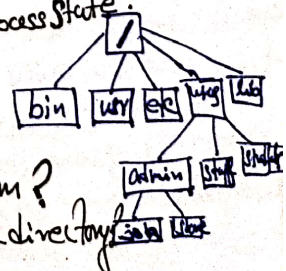
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Important Questions on UNIX By Subhadev Bhattacharjee [Asst. Professor of T.I.H.C.]

- Define OS. Write two parts of OS. Define multi programming and multitasking. "IS UNIX a multitasking OS"—Explain.
- What do the terms UID, PID and PPID stands for? For a currently executing process how can you obtain these values?
- Explain internal and external commands in UNIX. "UNIX is a multiuser System OS"—Explain. Write characteristics of UNIX O.S.
- Describe the layered architecture of UNIX OS. What are it's advantages?
- What are the different types of files available in UNIX?
- What command would you give read, write and execute permission, group members read and execute permission and give others just read permission for the file mid. (\$ chmod 754 mid)
- Describe link and symbolic link with utilities. Why the symbolic link of directory is two? What is race condition?
- Describe different kinds of files in UNIX. (Regular, Directory, Device file)
- Differentiate b/w absolute and relative path name.
- Differentiate between hard link and soft link.
- Explain the significance of file permissions for a directory.
- What is umask? How can file permissions be defined with it?
- Write down the applications of chown and chgrp commands.
- Define Absolute Path (A relative path describes the location of a file in relation to another file on the same current directory. Example: images/image123.gif)
- Define Relative Path (A path that contains the root directory and all other sub-directories required to get into the directory you are currently in or wish to get to. Example: /home/user/MyPicture/images/image123.gif)
- What are the utilities of environment variables?
- What are links? Differentiate b/w hard links and symbolic links. Differentiate b/w absolute path and relative path name.
- Short Note: Kernel, Shell, System Calls, Process State.
- Explain the structure of UNIX file system.
- How is copying a file different from linking it? Give suitable examples.
- What are the different blocks in UNIX file system? Explain briefly. How does UNIX access a file from a directory? Explain with example.
- What are the two ways to change the permissions associated with a file? (Symbolic permissions, Absolute Permissions)
- Explain with suitable example—"Directory permission is more important than file permission" (\$ chmod a-rwx abc)

P.T.O.

① directory →



23. What is filter? Give example. (When a program takes its i/p from another program, performs operations on that i/p and writes the results to the standard o/p, is called filter. Ex. Grep is used to search a file or pattern. SED and AWK are powerful filter in UNIX OS that consists of a sequence of pattern-action statements and optional function definitions.)

Example, `$cat apple.txt`

```
core
worm seed
jewel
$cat apple.txt | wc
```

output: 3 4 21
lines words characters

LAST 3 years 80% Must
Complete for Betterment

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24. What is escape sequence? Give Example.
25. Explain meaning of grep, egrep, fgrep.
(grep → basic regular expression, egrep → Uses extended regular expressions slowest, fgrep → No regular expressions fastest)
26. What is i-node? What are the various types of information stored in it? Explain: Date, LS, PS, CP, MV, CAT, Who, CHMOD, Pipe, grep.
27. What is the difference b/w the following vi commands?
:w ⇒ :w is to save the edited file with a new name.
:q! ⇒ :q! is used for quit vi without writing.
:wq ⇒ :wq is used to save the edited file and quit vi.
ZZ ⇒ ZZ is used for saving and quitting Vi.
Write if the buffer has been changed.
28. Write short note on Ex mode operations of vi editor.
29. What are the three modes of operation of vi?
Which one is the default mode?
30. What is Zombie Process? When does it occurs? Define Orphan Process.
31. Define sticky bit, daemon, nice command, touch command, nohup us. &, init process, PIP vs. PPID, Process State Diagram, exec system calls in process creation (Fork, Exec, Wait) IFC.
32. How a process is created? Mention briefly the role of the fork and exec system calls in process creation (Fork, Exec, Wait)
33. Explain the use of following UNIX commands: mv, cp, ps, top, du, gawk, PATH, TERM, PS1, PS2, HOME, nice, grep, sed, tar, nohup, who, wc, head, tail, &
34. Types of shell & it's features. Short note on bsh & csh, Command interpreter. ★ Must remember (10f) Command notes (marked) on UNIX.
35. What do you mean by positional parameter? How do you refer positional parameter within shell scripts? Give suitable example.
36. Write meaning of shell parameters: \$#, \$*, \$@, \$?, \$!
37. Why shell is called "command interpreter"?
38. WASS to check Prime, Palindrome, positive & negative, factorial, Fibonacci, Armstrong no, 1&2&3 combinations.